Take-Home Assignment

AI Production Cost Tracker

Software Engineer — Internal Tools & Automation

1. Background

Our production team generates AI images and videos using several third-party tools. At the end of every week, the operations team manually compiles spending per project from multiple sources. The process takes around three hours, is prone to errors, and budget overruns are usually discovered too late to act on.

2. Objective

Build, from scratch, a small internal tool that replaces this manual process. It should consolidate usage data from every source into one place, show the operations team where the money is going, and warn them before a project runs over budget.

3. Provided Materials

Item	Description
`vendor_a.csv`	Usage export from Vendor A. Amounts are in credits.
`vendor_b.csv`	Usage export from Vendor B. Amounts are in USD.
Vendor C API	A mock HTTP API for Vendor C usage data, in `vendor-c-mock/` (requires Node.js 18+ to run). See `docs/vendor_c_api.md` for endpoints, authentication and response format.
`manual_ops.xlsx`	The spreadsheet the operations team maintains by hand. Amounts are in IDR.
`projects.csv`	Active projects with their project codes and budgets (IDR).
`rates.json`	Fixed conversion rates for credits to IDR and USD to IDR.

4. Requirements

4.1 Data Import

- Import usage data from all four sources (Vendor A, Vendor B, Vendor C API and the manual spreadsheet) into a single database.
- Normalize every record: amounts converted to IDR using the provided rates, and timestamps expressed in WIB (UTC+7).
- Link each usage record to a project defined in `projects.csv`.
- Running the import again must not create duplicate records or inflate totals.

4.2 Dashboard

A web interface that the non-technical operations team can use on their own, showing:

- Spend versus budget for each project, including the percentage of budget used.
- Spend broken down by tool / vendor.
- Weekly spend over time.

4.3 Budget Alerts

- When a project's spend crosses **80% of its budget**, send a notification to a webhook (Discord, Telegram, Slack, or a mock endpoint of your choice).
- The message should state the project, current spend, budget and percentage used.

4.4 General

- Where a requirement is unclear, make a reasonable decision and document it.
- Keep credentials and secrets out of the codebase.
- Include tests for the logic you consider most critical.

5. Deliverables

- A link to your Git repository, with the commit history preserved.
- A **README** covering:
  - How to set up and run the project, including the import and the alerts.
  - Architecture overview and key design decisions.
  - Assumptions you made.
  - Known limitations and what you would do next with more time.